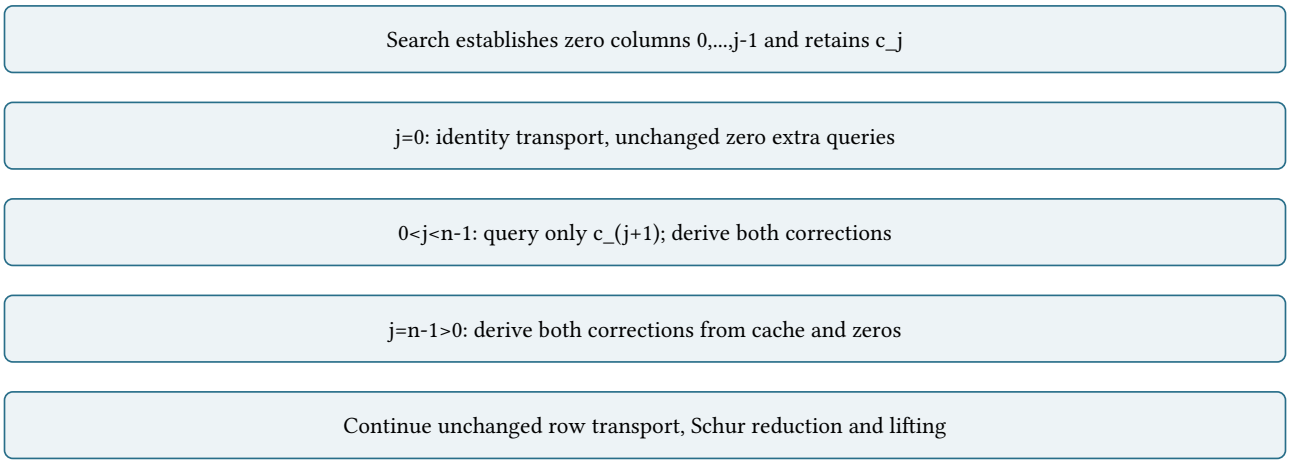
# Reuse pivot-search queries in column transport

2026-11-02 · Derived operation schedule · Syntax/static checks only

**Construction.** The scalar-pivot search already queries columns until the first nonzero c_j=Ae_j. Retain that column and the established zero prefix to avoid duplicate work in the two-query column-swap primitive.




The three middle boxes are mutually exclusive branches. All continue to the same row-transport and solve/witness path.

**Identity.** For j>0 and u=e_0−e_j, A u=−c_j. The second generator correction −A Z_b u equals c_(j+1)−c_1 for an interior column and −c_1+b c_0 for the last column. Here c_0=0, and c_1 is cached when j=1 or already known zero when j>1. Thus the interior branch needs only the following column; the last-column branch needs no query. For n=2,j=1, both corrections are the same cached negative column.

**Derived transport counts, excluding query internals.**

| Branch | Old queries | New queries | New products | New add/sub |
|---|---|---|---|---|
| j=0 | 0 | 0 | 0 | 0 |
| j=1 interior | 2 | 1 | 0 | 2n+1 |
| j>1 interior | 2 | 1 | 0 | n+1 |
| Last j>0 | 2 | 0 | 0 | n+1 |

For every old nonidentity branch, explicit scalar work was two products and n+1 add/sub operations. The first interior branch now adds n subtractions. Search query count and entry-test order are unchanged. Copying the canonical wrap b eliminates multiplication by one. Generator width remains s+2, including a zero correction when applicable.

**Integration and provenance.** A separate solve_cached.py changes only the search return and transport call, plus one import. Normalized AST comparison checks every other executable statement against the preserved original solver. The internal cache is tied to the source object/signature and is consumed once. It retains one length-n vector. In-place source mutation and forged cache objects are outside this internal contract; the cache is not an independently verified public certificate.

**Scope.** Nonsingular nodes have j=0, so this optimization targets singular intermediate systems with zero-prefix columns. Query vectors differ from the reference: native bit-length volumes and runtime must be evaluated, not inferred from call counts. No numerical or performance experiment ran.

**Next checks.** All branch boundaries, n=2, wrap values, pivot rows, wrong-source/reuse/malformed cache controls, exact scalar/query counters, reference generator equality and both solve/witness outcomes. Preserve the full original cohort; charge cached vectors and audit storage.

**Sources.** cached_transport.py, solve_cached.py, check_preparation.py and DERIVATION.md; original companion-rank-solver column-swap identity. Bostan et al. (2017), Section 6, DOI 10.1137/16M1062855 supplies prior structured-solve context; no randomized or fast bound is imported.